1. Problem Statement

The BIS Compliance Gap for Indian MSEs

- India has 63+ million Micro & Small Enterprises (MSEs) in manufacturing
- BIS SP 21 contains hundreds of standards across building materials — cement, steel, concrete, aggregates
- MSEs spend weeks manually searching applicable BIS standards before product launch
- Errors in compliance identification lead to legal penalties, market delays & lost business
- No affordable, automated tool exists to match product descriptions to relevant BIS standards

Challenge: Turn a product description into accurate BIS standard recommendations — in seconds.

2. Solution Overview

BISense AI — AI-Powered BIS Standards Recommendation Engine

- Accepts a natural-language product description as input
- Runs a RAG (Retrieval-Augmented Generation) pipeline over BIS SP 21 documents
- Returns top 3–5 most relevant BIS standards with brief rationale for each
- Deployed as a full-stack web application at bisense-ai-2026.web.app
- Covers Building Materials category: Cement, Steel, Concrete, Aggregates, Bricks & more
- Inference entry point: python inference.py --input dataset.json --output results.json

Stack: Python · FAISS · Sentence Transformers · Gemini 2.0 Flash · Firebase Hosting · React + Vite frontend

3. System Architecture

FIREBASE HOSTING (CDN) · bisense-ai-2026.web.app

React + Vite Frontend

Landing Page	Compliance Check	Analytics Rule Book	Standards Browser
Dashboard View	Compliance Graph	History Page	Chatbot (Voice AI)

api

GOOGLE CLOUD RUN · Backend

FastAPI Application

/api/compliance/search → Hybrid Retrieval (BM25 + Dense)
/api/chat → Vertex AI Gemini 2.0 Flash
/api/voice/tts → Google Cloud Text-to-Speech
/api/analytics → Metrics Engine
/api/standards → Registry Browser (496 Standards)

BIS Registry 496 Standards	**Translator** 4 Languages	**Gemini AI** 2.0 Flash

4. Chunking & Retrieval Strategy

Document Ingestion & Chunking

- Source: BIS SP 21 raw PDFs (Building Materials section)
- Parsed with pdfplumber / PyMuPDF to extract standard summaries
- Semantic chunking by standard number — each BIS standard = one chunk
- Chunk size: ~200–400 tokens preserving IS number, title & summary
- Metadata preserved: Standard ID, category, material type

Embedding & Retrieval

- Embedding model: sentence-transformers/all-MiniLM-L6-v2
- Hybrid retrieval: BM25 (keyword) + Dense (semantic) combined
- Vector store: FAISS IndexFlatL2 for fast nearest neighbour search
- Query expansion: rephrase product description for better match
- Top-K retrieval: fetch top 10, re-rank with cross-encoder
- LLM synthesis: top 3–5 chunks with rationale via Gemini 2.0 Flash

5. Demo Highlights

Live at: bisense-ai-2026.web.app

Example Query: "OPC 53-grade cement bags for structural concrete in high-rise buildings"

Returned Recommendations:

IS 269:2015	Ordinary Portland Cement — Specification (53 Grade)
IS 456:2000	Plain & Reinforced Concrete — Code of Practice
IS 516:1959	Methods of Tests for Strength of Concrete
IS 4031:1988	Methods of Physical Tests for Hydraulic Cement

- Each result includes a one-line rationale explaining relevance to the query
- Average response time: ~2–3 seconds end-to-end | Voice AI (TTS) available in Chatbot

6. Evaluation Results

≥ 80%	≥ 0.70	< 3 sec
Hit Rate @3	MRR @5	Avg Latency

Public Test Set Results (10 Sample Queries)

Metric	Target	Achieved	Status
Hit Rate @3	> 80%	100.00%	PASS
MRR @5	> 0.70	1.0000	PASS
Avg Latency (sec)	< 5.0	0.0004s	PASS
Hallucination Rate	0%	0%	PASS
Relevance Score	High (1-5)	5/5	PASS

All standards returned are genuine BIS IS numbers — zero hallucinations on test set.

7. Impact on MSEs

Transforming BIS Compliance for Small Businesses

Before BISense AI

- Weeks of manual BIS document research
- Expensive consultants for compliance mapping
- Risk of missing applicable standards
- Delayed product launches & certifications
- Inaccessible to micro enterprises with no legal team

After BISense AI

- Instant recommendations in under 5 seconds
- Zero cost — open web tool accessible to all MSEs
- Reduces compliance errors with AI-verified standards
- Faster product launch and ISI mark applications
- Scalable to cover all BIS SP 21 product categories

Target Beneficiaries: 63M+ Indian MSEs in manufacturing, with focus on Building Materials sector

8. Team & Acknowledgements

Team Kryptonite

Name	Role
Hardik Hinduja	Team Leader
Sahil Deshmukh	Member
Siddharth Dey	Member
Avinash Gehi	Member

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